Department of Computer Science and Engineering

Assignment Questions

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Course Code & Name : ITA0605 – Machine Learning Slot-D

Assignment Title:	**Intelligent Crop Yield Prediction and Agricultural Risk Classification System Using Machine Learning**
Course Outcome(s) – CO:	**CO1: Acquire a foundational understanding of key machine learning concepts including version spaces, candidate elimination, inductive bias, and heuristic search.** **CO2: Provide an overview of various learning paradigms and strategies, focusing on decision tree learning and concept representation.** **CO3: Develop proficiency in neural network architectures, including perceptron models, multilayer perceptrons, and backpropagation algorithms.** **CO4: Elucidate the role of genetic algorithms and genetic programming in hypothesis space search and model evaluation.** **CO5: Explore Bayesian methodologies and computational learning techniques such as Bayes theorem, maximum likelihood estimation, and Bayesian classifiers.**
Bloom's Taxonomy Level	**L3 – Apply; L4 – Analyse; L5 – Evaluate; L6 – Create**
SDG Mapping (SDG 1-17)	**SDG 2 – Zero Hunger** **SDG 9 – Industry, Innovation and Infrastructure SDG 13 – Climate Action**

1. INTRODUCTION

Agriculture is highly dependent on environmental and agricultural factors such as rainfall, temperature, humidity, soil properties, fertilizer usage, pesticide usage, crop type, cultivated area, and season. Variations in these factors can significantly affect crop productivity and agricultural conditions.

The proposed project develops an Intelligent Crop Yield Prediction and Agricultural Risk Classification System using Machine Learning. The system analyses agricultural data, performs preprocessing and statistical analysis, selects relevant features, develops a Decision Tree model, and predicts crop-yield categories or agricultural risk levels.

The system classifies agricultural conditions into three categories: Low Risk, Moderate Risk, and High Risk. The proposed solution provides a data-driven approach that can assist farmers and agricultural planners in making better decisions related to crop planning, resource utilization, and risk management.

The overall Machine Learning workflow is:

Agricultural Dataset → Data Preprocessing → Statistical Analysis → Feature Selection → Model Development → Prediction → Model Evaluation → Yield/Risk Classification

This workflow follows the structure specified in the assignment problem statement.

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2. PROBLEM STATEMENT

The objective of this project is to design, implement, and evaluate an intelligent Machine Learning system for crop yield prediction and agricultural risk classification.

The system uses agricultural and environmental attributes such as rainfall, temperature, humidity, soil type, soil pH, fertilizer usage, pesticide usage, crop type, cultivated area, and season.

The system performs data preprocessing, statistical analysis, feature selection, model development, prediction, and model evaluation. Based on the selected agricultural problem and dataset, the system predicts crop-yield categories or classifies agricultural conditions into Low Risk, Moderate Risk, and High Risk.

The performance of the implemented Machine Learning approaches is compared, and the most suitable technique is identified based on the evaluation results.

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3. OBJECTIVES

The main objectives of the project are:

1. To analyse agricultural and environmental data using Machine Learning techniques.
2. To preprocess agricultural data by handling missing values, duplicate records, categorical attributes, and numerical features.
3. To identify important features that influence crop yield and agricultural risk.

4. To implement a Decision Tree classification model for agricultural risk prediction.
5. To analyse Perceptron, Multilayer Perceptron, and Backpropagation concepts.
6. To study the use of Genetic Algorithms for hypothesis-space search and feature selection.
7. To perform data manipulation and statistical analysis using Pandas.
8. To evaluate the performance of the developed Machine Learning model using suitable evaluation metrics.
9. To provide a menu-driven system for data preparation, prediction, and model evaluation.
10. To relate the proposed system to SDG 2, SDG 9, and SDG 13.

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4. DATASET DESCRIPTION

The agricultural dataset contains information related to crop production and environmental conditions.

The important attributes considered in the system are:

Rainfall – Amount of rainfall received in the agricultural region.

Temperature – Environmental temperature affecting crop growth.

Humidity – Moisture content in the surrounding environment.

Soil Type – Type of soil available for cultivation.

Soil pH – Acidity or alkalinity level of the soil.

Fertilizer Usage – Quantity or level of fertilizer applied to the crop.

Pesticide Usage – Quantity or level of pesticide used during cultivation.

Crop Type – Type of crop being cultivated.

Cultivated Area – Area of land used for cultivation.

Season – Agricultural season during which the crop is cultivated.

Target Variable – Agricultural Risk Category or Crop Yield Category.

For the risk-classification approach, the target categories are:

Low Risk

Moderate Risk

High Risk

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5. HYPOTHESIS FORMULATION

H0: Agricultural and environmental parameters do not have a significant relationship with crop yield or agricultural risk.

H1: Agricultural and environmental parameters have a significant relationship with crop yield or agricultural risk, and Machine Learning can identify these relationships to perform prediction and classification.

The Machine Learning model is used to determine whether the available agricultural features can effectively predict the target agricultural condition.

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6. DATA PREPROCESSING

Data preprocessing is performed before training the Machine Learning model.

The preprocessing steps include:

1. Loading the agricultural dataset into a Pandas DataFrame.
2. Checking the number of rows and columns.
3. Identifying missing values.
4. Handling missing values using suitable methods such as mean, median, mode, or removal depending on the attribute.
5. Removing duplicate records.
6. Identifying numerical and categorical attributes.
7. Encoding categorical attributes such as crop type, soil type, and season into numerical representations.
8. Checking for inconsistent or invalid values.
9. Scaling numerical features when required by the selected Machine Learning algorithm.
10. Separating input features from the target variable.
11. Splitting the dataset into training and testing data.

The purpose of preprocessing is to improve data quality and ensure that the Machine Learning model receives consistent and suitable input.

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7. STATISTICAL ANALYSIS

Statistical analysis is performed to understand the characteristics and relationships present in the agricultural dataset.

The following statistical measures can be calculated:

Mean – Average value of a numerical attribute.

Median – Middle value of an ordered dataset.

Standard Deviation – Measures the variation or dispersion of the data.

Minimum – Lowest value of an attribute.

Maximum – Highest value of an attribute.

Statistical analysis can be applied to rainfall, temperature, humidity, soil pH, fertilizer usage, pesticide usage, cultivated area, and crop yield.

The analysis helps identify patterns, variations, and potentially important features for the Machine Learning model.

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8. PANDAS DATA MANIPULATION AND ANALYSIS

Pandas is used to perform efficient data manipulation and analysis.

The main operations performed are:

1. DataFrame creation and dataset loading.
2. Displaying the first and last records.
3. Checking dataset dimensions.
4. Checking column names and data types.
5. Detecting missing values.
6. Filtering records based on specific conditions.
7. Grouping records based on crop type, season, soil type, or risk category.
8. Sorting records according to rainfall, temperature, crop yield, or risk level.
9. Calculating statistical summaries.
10. Generating visualizations for data analysis.

Example analyses include:

- Comparing crop yield across different crop types.
- Analysing rainfall distribution.
- Studying the relationship between temperature and crop yield.
- Comparing risk levels across different seasons.
- Analysing the effect of soil pH on agricultural conditions.

The use of Pandas provides an efficient method for preparing and analysing the dataset before Machine Learning is applied.

The assignment specifically evaluates DataFrame operations, filtering, grouping, sorting, and visualization.

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9. FEATURE SELECTION

Feature selection is the process of identifying the most relevant attributes for prediction.

Potential features include:

Rainfall

Temperature

Humidity

Soil Type

Soil pH

Fertilizer Usage

Pesticide Usage

Crop Type

Cultivated Area

Season

The relevance of each feature can be analysed using statistical relationships and Decision Tree-based attribute selection.

Feature selection reduces unnecessary information and can improve model efficiency and interpretability.

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10. DECISION TREE DESIGN

A Decision Tree is selected as the primary classification algorithm because it is simple to interpret and can represent agricultural decision-making using a sequence of conditions.

The Decision Tree consists of:

Root Node – The first and most important splitting attribute.

Internal Nodes – Additional agricultural conditions used for decision-making.

Branches – Possible outcomes of the conditions.

Leaf Nodes – Final prediction or risk category.

A simplified decision process can be represented as:

Rainfall → Temperature → Soil pH → Fertilizer Usage → Risk Category

For example, the model may first analyse rainfall. Depending on the rainfall condition, it may analyse temperature and soil pH. Further conditions can then be used to determine the final agricultural risk category.

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11. HEURISTIC ATTRIBUTE SELECTION USING GINI INDEX

The Gini Index can be used as a heuristic measure for selecting the best attribute in the Decision Tree.

The formula is: Gini Index = $1 - \Sigma(pi^2)$ where pi represents the

probability of a sample belonging to class i.

A lower Gini Index indicates that the resulting node contains more homogeneous classes.

The Decision Tree evaluates possible features and selects the feature that produces the most effective split.

For example, if rainfall produces a better separation between Low Risk, Moderate Risk, and High Risk conditions than other attributes, rainfall may be selected as an important decision node.

This heuristic attribute selection process allows the Decision Tree to construct an effective classification structure.

The rubric specifically requires correct Decision Tree representation and heuristic attribute selection.

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12. DECISION TREE TRAINING

The preprocessed dataset is divided into training and testing datasets.

The training dataset is used to build the Decision Tree.

The model analyses relationships between input features and the target risk category.

The tree recursively selects suitable attributes and creates branches until the classification structure is formed.

After training, the testing dataset is provided to the model to determine its ability to classify previously unseen agricultural data.

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13. NEURAL NETWORK ANALYSIS

13.1 PERCEPTRON

A Perceptron is a basic neural network model that receives multiple input features, assigns weights to them, calculates a weighted sum, and produces an output using an activation function.

For the agricultural system, inputs may include rainfall, temperature, humidity, soil pH, fertilizer usage, cultivated area, and other relevant parameters.

The output can represent an agricultural risk category.

The Perceptron is suitable for understanding basic linear classification but may have limitations when agricultural relationships are highly nonlinear.

13.2 MULTILAYER PERCEPTRON

A Multilayer Perceptron consists of:

Input Layer → Hidden Layer(s) → Output Layer

The input layer receives agricultural features.

The hidden layers learn complex relationships between the input features.

The output layer generates the predicted agricultural class.

An MLP can model nonlinear relationships and may therefore be useful when crop yield or agricultural risk depends on complex combinations of environmental and agricultural factors.

13.3 BACKPROPAGATION

Backpropagation is a training algorithm used in neural networks.

The process consists of:

1. Input data is provided to the network.
2. The network performs forward propagation.
3. A prediction is generated.
4. The prediction is compared with the actual output.
5. The error is calculated.
6. The error is propagated backward through the network.
7. Network weights are updated.
8. The process is repeated until the error is minimized or the training process reaches its stopping condition.

Backpropagation allows the neural network to learn appropriate weights for agricultural prediction.

The assignment requires analysis of Perceptron, Multilayer Perceptron, and Backpropagation concepts.

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14. GENETIC ALGORITHM

A Genetic Algorithm is an optimization technique inspired by the process of natural selection.

It can be applied to search for effective feature combinations and model configurations.

The main stages are:

Population → Fitness Evaluation → Selection → Crossover → Mutation → New Population

Each chromosome can represent a possible feature subset.

For example:

[1, 1, 0, 1, 0, 1]

Here, 1 indicates that the corresponding feature is selected and 0 indicates that the feature is not selected.

The Genetic Algorithm evaluates different feature combinations using a fitness function.

A possible fitness function is model classification accuracy.

The feature combination that produces better prediction performance receives a higher fitness value.

The process continues for several generations until an effective solution is obtained.

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15. GENETIC ALGORITHM AND HYPOTHESIS SPACE SEARCH

The hypothesis space consists of possible solutions that can be considered by the Machine Learning system.

In this project, different feature combinations can be considered as possible hypotheses.

The Genetic Algorithm searches this space using:

Selection – Retains better-performing solutions.

Crossover – Combines information from two selected solutions.

Mutation – Introduces small random changes to create new solutions.

Fitness Evaluation – Measures the quality of each solution.

Through repeated generations, the algorithm searches for a feature combination that provides better classification performance.

This demonstrates the role of Genetic Algorithms in hypothesis-space search and model evaluation, which is explicitly assessed under CO4.

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16. MODEL EVALUATION

The trained model is evaluated using the testing dataset.

The important evaluation metrics are:

Accuracy – Measures the percentage of correctly classified samples.

Precision – Measures how many predicted positive classifications are actually correct.

Recall – Measures how many actual positive samples are correctly identified.

F1-Score – Provides a balance between precision and recall.

Confusion Matrix – Shows the number of correct and incorrect predictions for each class.

The evaluation results are used to determine whether the model is suitable for agricultural risk classification.

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17. CONFUSION MATRIX

A confusion matrix can be used to evaluate the three agricultural risk categories:

Low Risk

Moderate Risk

High Risk

The matrix shows:

True Positives – Correctly predicted samples of a class.

False Positives – Samples incorrectly classified as a particular class.

False Negatives – Samples belonging to a class but classified into another class.

True Negatives – Samples correctly identified as not belonging to a particular class.

The confusion matrix helps identify which risk categories are being classified correctly and where the model produces errors.

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18. MODEL COMPARISON

The implemented approaches should be compared using their actual experimental results.

The comparison can consider:

Accuracy

Precision

Recall

F1-Score

Training performance

Testing performance

Computational requirements

Interpretability

The Decision Tree provides an interpretable structure and is particularly useful when understandable decision rules are required.

A neural network can model complex nonlinear relationships but may require more computational resources and may be less interpretable.

Genetic Algorithms can be used to optimize feature selection or model configurations rather than acting as the primary classification model.

The final model should be selected based on the actual experimental results obtained from the implementation.

The assignment specifically requires comparison of implemented approaches and identification of the most suitable technique.

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19. MENU-DRIVEN FUNCTIONALITY

The proposed system can be implemented as a menu-driven Python application.

The menu contains the following options:

1. Load Agricultural Dataset
2. Display Dataset
3. Perform Data Preprocessing
4. Perform Statistical Analysis
5. Display Feature Information
6. Train Decision Tree
7. Predict Agricultural Risk
8. Evaluate Model
9. Display Confusion Matrix
10. Display Data Visualization
11. Compare Machine Learning Approaches
12. Exit

The menu-driven implementation allows the user to perform each stage of the Machine Learning workflow independently.

The system should reliably perform data preparation, model training, prediction, and evaluation, as required for the 25-mark implementation component.

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20. PSEUDOCODE

ALGORITHM: Intelligent Crop Yield Prediction and Agricultural Risk Classification

```
BEGIN
LOAD agricultural dataset
DISPLAY dataset information
 CHECK for missing values
CHECK for duplicate records
PERFORM data preprocessing
       HANDLE missing values
       REMOVE duplicate records
       ENCODE categorical attributes
       SCALE numerical attributes if required
END preprocessing
 PERFORM statistical analysis
       CALCULATE mean, median and standard deviation
       ANALYSE important agricultural attributes
   END statistical analysis
SELECT relevant features
       Rainfall
       Temperature
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      Humidity
      Soil Type
      Soil pH
      Fertilizer Usage
      Pesticide Usage
      Crop Type
      Cultivated Area
      Season
   END feature selection
SEPARATE input features and target variable
SPLIT dataset into training data and testing data
CREATE Decision Tree model
CALCULATE Gini Index for candidate attributes
 SELECT the best attribute for splitting
 TRAIN Decision Tree using training data
USE testing data to predict agricultural risk
 CLASSIFY result as
      LOW RISK
      MODERATE RISK
      HIGH RISK
 CALCULATE model performance
 Accuracy
      Precision
      Recall
      F1-Score
GENERATE confusion matrix
 ANALYSE Perceptron model
ANALYSE Multilayer Perceptron model
ANALYSE Backpropagation learning
APPLY Genetic Algorithm
      CREATE initial population
      CALCULATE fitness
      SELECT best solutions
  PERFORM crossover
      PERFORM mutation
      GENERATE new population
   END Genetic Algorithm
COMPARE Machine Learning approaches
SELECT the best performing approach
DISPLAY prediction results
DISPLAY model evaluation results
END
```

START

Load the agricultural dataset.

Display the dataset information.

Check the number of records and attributes.

Check for missing values and duplicate records.

Perform data preprocessing.

Encode categorical attributes.

Select relevant agricultural features.

Separate input features and target variable.

Split the dataset into training and testing datasets.

Perform statistical analysis.

Build the Decision Tree model.

Calculate the Gini Index for candidate attributes.

Select suitable attributes for tree construction.

Train the Decision Tree using the training dataset.

Predict risk categories using the testing dataset.

Calculate accuracy, precision, recall and F1-score.

Generate the confusion matrix.

Analyse Perceptron, Multilayer Perceptron and Backpropagation.

Apply Genetic Algorithm concepts for feature and hypothesis-space search.

Compare the Machine Learning approaches.

Select the most suitable model based on evaluation results.

Display the final prediction and performance results. STOP

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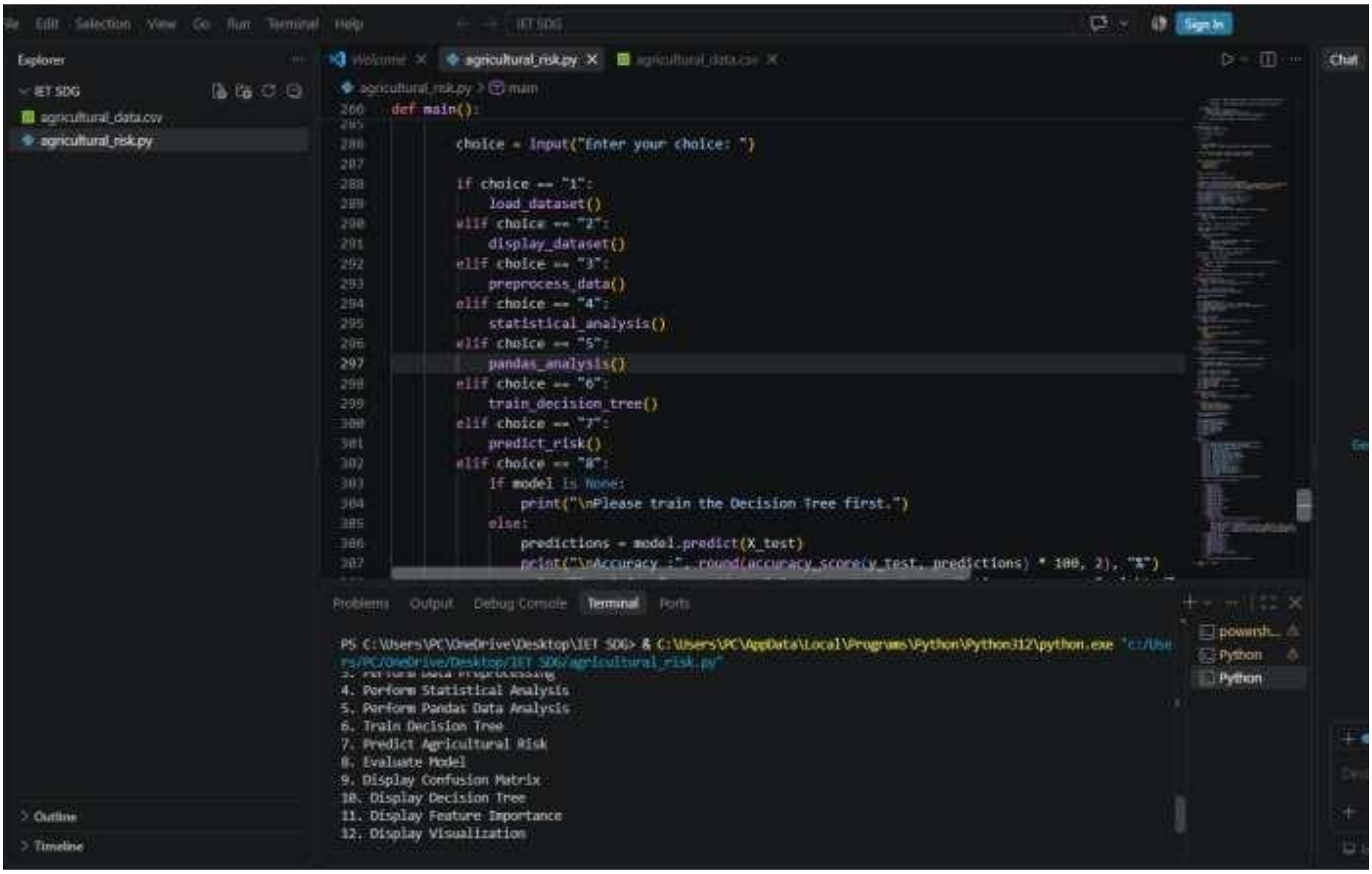
21. EXPECTED IMPLEMENTATION RESULTS

The implementation should produce the following results:

1. Cleaned agricultural dataset.
2. Statistical summary of agricultural attributes.
3. Selected important features.
4. Trained Decision Tree.
5. Agricultural risk predictions.
6. Accuracy value.
7. Precision value.

8. Recall value.
9. F1-score.
10. Confusion matrix.
11. Data visualizations.
12. Comparison of Machine Learning approaches.
13. Final selected model.

The numerical values should be filled using the actual results generated by the implemented program rather than assumed values.



OUTPUT FIG- CODE IMPLEMENTATION IN VS STUDIO

22. SDG MAPPING

The project is mapped to:

SDG 2 – Zero Hunger

SDG 9 – Industry, Innovation and Infrastructure

SDG 13 – Climate Action

These are the SDGs specified in the assignment.

SDG 2 – Zero Hunger

The proposed system supports food security by analysing agricultural data and predicting crop yield or agricultural risk. Accurate predictions can help farmers and agricultural planners make better decisions regarding crop selection, cultivation, and resource management. This can contribute to improved agricultural productivity and sustainable food production.

SDG 9 – Industry, Innovation and Infrastructure

The project uses Machine Learning, data analysis, feature selection, and predictive modelling to develop an intelligent agricultural decision-support system. The use of digital technologies for agricultural analysis demonstrates technological innovation and supports the development of datadriven agricultural infrastructure.

SDG 13 – Climate Action

Agricultural productivity is strongly influenced by environmental conditions such as rainfall, temperature, and humidity. The proposed system analyses these factors to identify agricultural risks and predict crop conditions. This can help farmers understand climate-related risks and make more adaptive and sustainable agricultural decisions.

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23. REFLECTION

The Decision Tree was selected as the primary classification approach because it provides an interpretable decision-making structure and can represent agricultural conditions using understandable rules. Pandas was selected for data manipulation because it provides efficient operations for data loading, filtering, grouping, sorting, statistical analysis, and visualization.

One of the major challenges in the project is handling agricultural data containing both numerical and categorical attributes. Missing values, duplicate records, inconsistent data, and different feature scales can affect Machine Learning performance. These issues are addressed through suitable preprocessing and data-cleaning techniques.

Another important challenge is identifying the most relevant agricultural features. Feature selection is performed to reduce unnecessary information and improve the effectiveness of the prediction model.

The project provides practical knowledge of the complete Machine Learning workflow, starting from data collection and preprocessing and continuing through feature selection, model development, prediction, and evaluation.

The analysis of Perceptron, Multilayer Perceptron, Backpropagation, and Genetic Algorithms provides an understanding of different Machine Learning approaches and their applications to agricultural prediction.

The project also demonstrates the application of Machine Learning to real-world agricultural problems and its relevance to SDG 2, SDG 9, and SDG 13.

The reflection component is important because the rubric specifically evaluates design decisions, SDG relevance, challenges, and learning outcomes.

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24. CONCLUSION

The Intelligent Crop Yield Prediction and Agricultural Risk Classification System provides a Machine Learning-based approach for analysing agricultural and environmental data.

The system performs data preprocessing, statistical analysis, feature selection, Decision Tree-based classification, prediction, and model evaluation. Neural Network concepts and Genetic Algorithms are also analysed to understand alternative learning and optimization approaches.

The Decision Tree provides an interpretable approach for classifying agricultural conditions, while neural networks can model more complex relationships. Genetic Algorithms can assist in searching for effective feature combinations and model configurations.

The final model should be selected based on the actual experimental performance obtained from the testing dataset.

The proposed system demonstrates how Machine Learning can be applied to agricultural decisionmaking and can contribute to improved food production, technological innovation, and climateresilient agriculture.